

MOSFETs

Metal-oxide-semiconductor
field-effect transistors

Lecture 3

Dr. Zahraa Ismail

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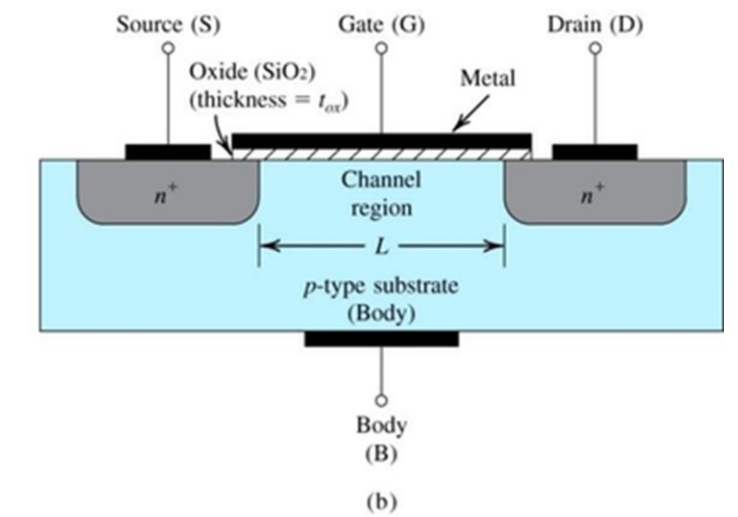
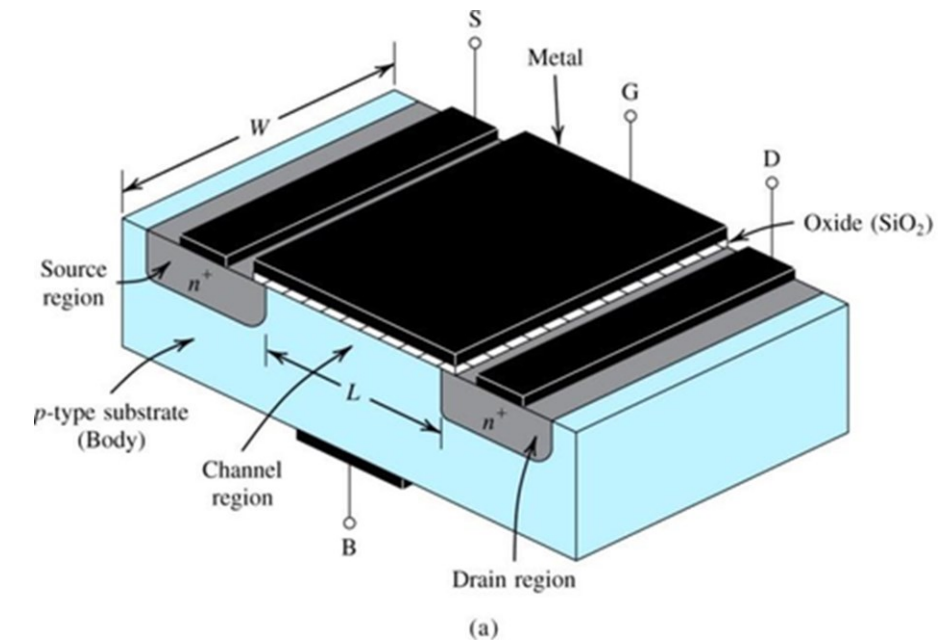


Figure: Sedra & Smith, Microelectronic Circuits

Lecture map

01 Structure and operation

how a channel forms; the threshold voltage

02 I–V characteristics

cutoff, triode and saturation

03 MOSFETs in DC circuits

finding the operating point

04 The MOSFET as an amplifier

transconductance, small-signal model

05 The MOSFET as a switch

BY THE END YOU CAN

- Explain how v_{GS} creates a channel and how v_{DS} pinches it off
- Pick the right region and use its i_D equation
- Find the DC operating point of a MOSFET circuit

02

I–V characteristics

Increasing v_{DS} : pinch-off

Triode and saturation

The i_D – v_{DS} family of curves

1 Structure and operation

2 I–V characteristics

3 MOSFETs in DC circuits

4 The MOSFET as an amplifier

5 The MOSFET as a switch

Creating a channel: the threshold voltage

A positive v_{GS} pushes holes away from the region under the gate and draws electrons in from the n^+ source and drain.

Once enough electrons gather, they form an n -type **inversion layer**: the channel that links source to drain.

DEFINITION

Threshold voltage V_t : the value of v_{GS} at which the channel just forms.

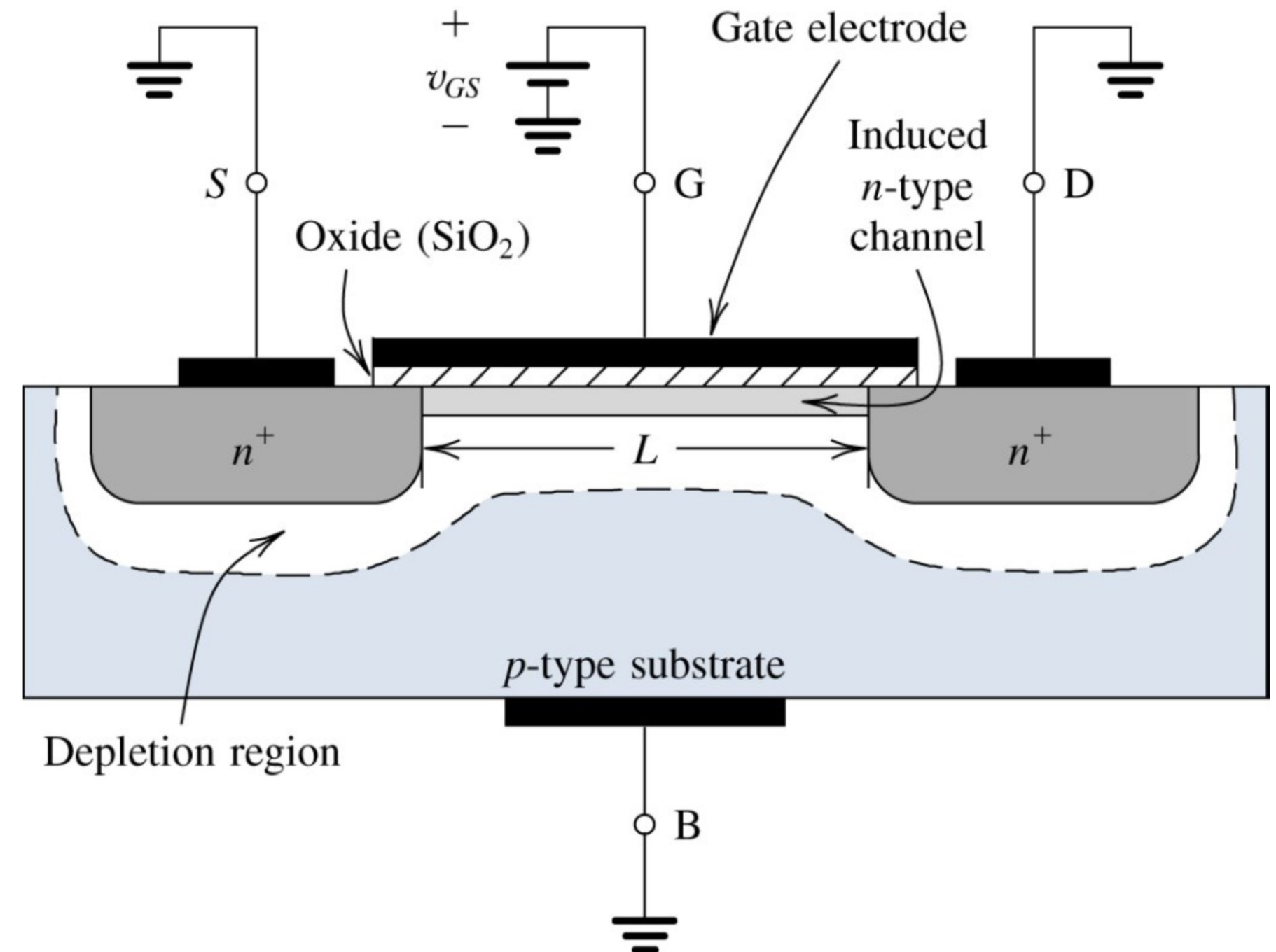
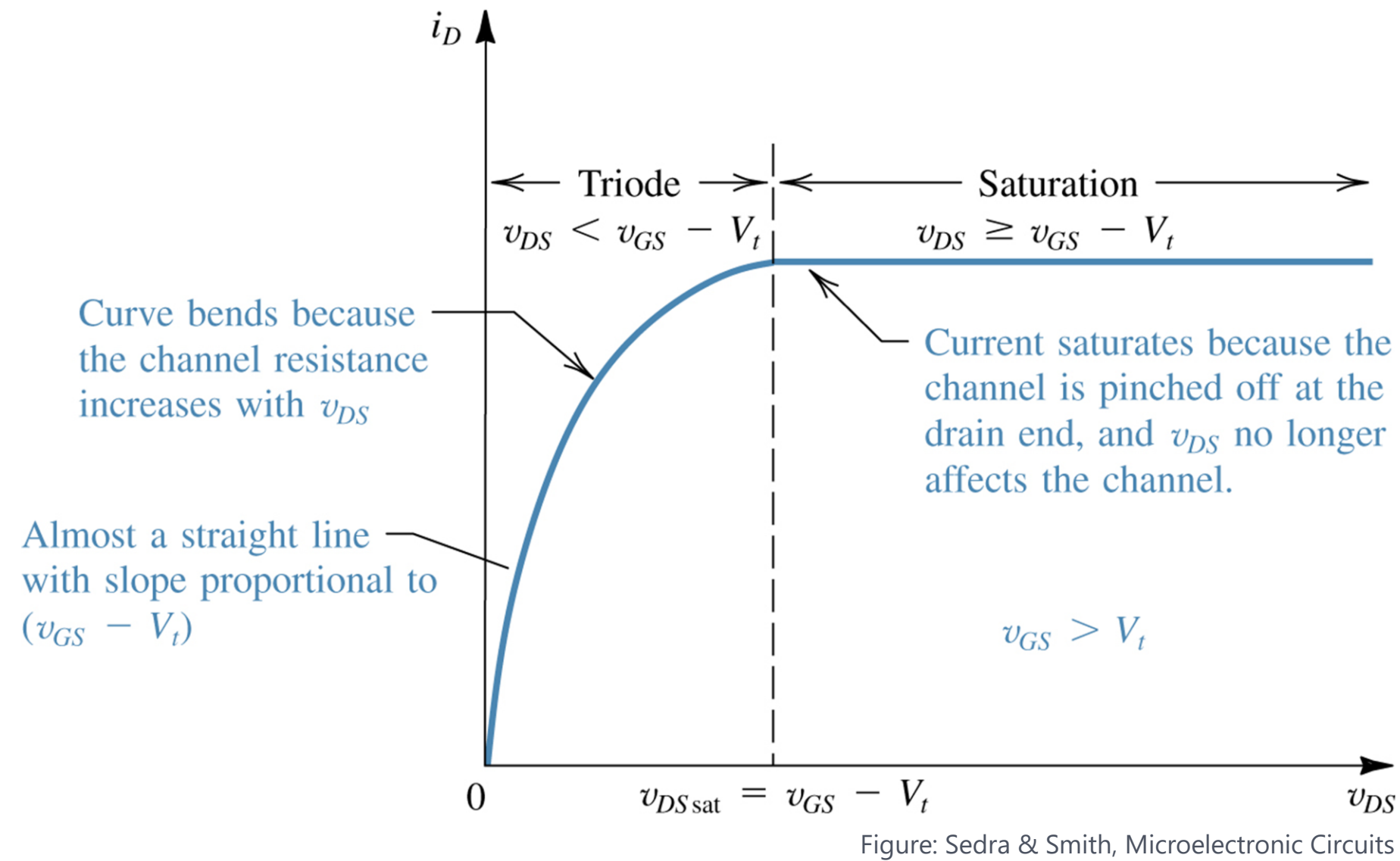


Figure: Sedra & Smith, Microelectronic Circuits

Triode and saturation: one curve, two equations



Triode $v_{DS} < v_{GS} - V_t$

$$i_D = k'_n \frac{W}{L} \left[(v_{GS} - V_t) v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$

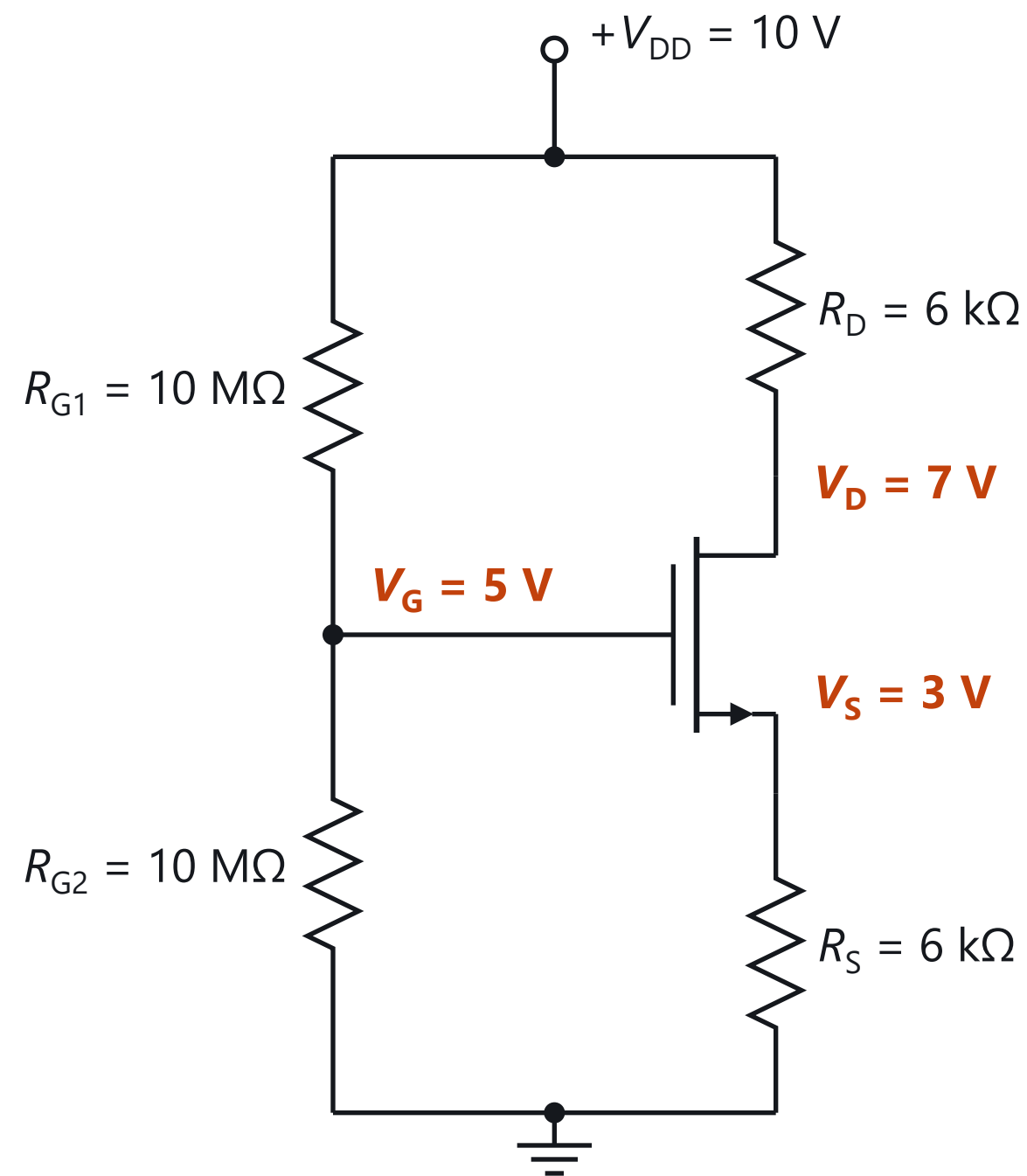
Saturation $v_{DS} \geq v_{GS} - V_t$

$$i_D = \frac{1}{2} k'_n \frac{W}{L} (v_{GS} - V_t)^2$$

Both require $v_{GS} > V_t$: without a channel, $i_D = 0$ (cutoff).

Example: find the drain voltage

Given $V_t = 1 \text{ V}$ and $k'_n(W/L) = 1 \text{ mA/V}^2$. Assume the gate draws no current.



- 1 Gate voltage: the divider is unloaded

$$V_G = V_{DD} \frac{R_{G2}}{R_{G1} + R_{G2}} = 5 \text{ V}$$

- 2 Assume saturation, with $V_{GS} = V_G - I_D R_S$ (I in mA, R in $\text{k}\Omega$)

$$I_D = \frac{1}{2} (1)(5 - 6I_D - 1)^2$$

- 3 Solve the quadratic

$$18I_D^2 - 25I_D + 8 = 0 \Rightarrow I_D = 0.89 \text{ or } 0.5 \text{ mA}$$

- 4 $I_D = 0.89 \text{ mA}$ would give $V_{GS} = -0.33 \text{ V} < V_t$ (no channel): reject it. So $I_D = 0.5 \text{ mA}$, $V_{GS} = 2 \text{ V}$.

- 5 $V_D = V_{DD} - I_D R_D = 10 - 0.5(6) = 7 \text{ V}$

$$V_D = 7 \text{ V}$$

✓ Check: $V_{DS} = 4 \text{ V} \geq V_{GS} - V_t = 1 \text{ V}$, so saturation holds.

Summary: three regions of operation

REGION	CONDITION	DRAIN CURRENT
Cutoff	$V_{GS} \leq V_t$	$i_D = 0$
Triode	$V_{GS} > V_t, V_{DS} < V_{GS} - V_t$	$i_D = k'_n \frac{W}{L} \left[(v_{GS} - V_t)v_{DS} - \frac{1}{2}v_{DS}^2 \right]$
Saturation	$V_{GS} > V_t, V_{DS} \geq V_{GS} - V_t$	$i_D = \frac{1}{2}k'_n \frac{W}{L} (v_{GS} - V_t)^2$

- In a DC problem: **assume** a region, **solve**, then **check** the assumption.
- Current saturates because the channel is pinched off at the drain end.